

Salesforce Platform Architecture

How Salesforce helps building secure and scalable apps at Lightning Speed

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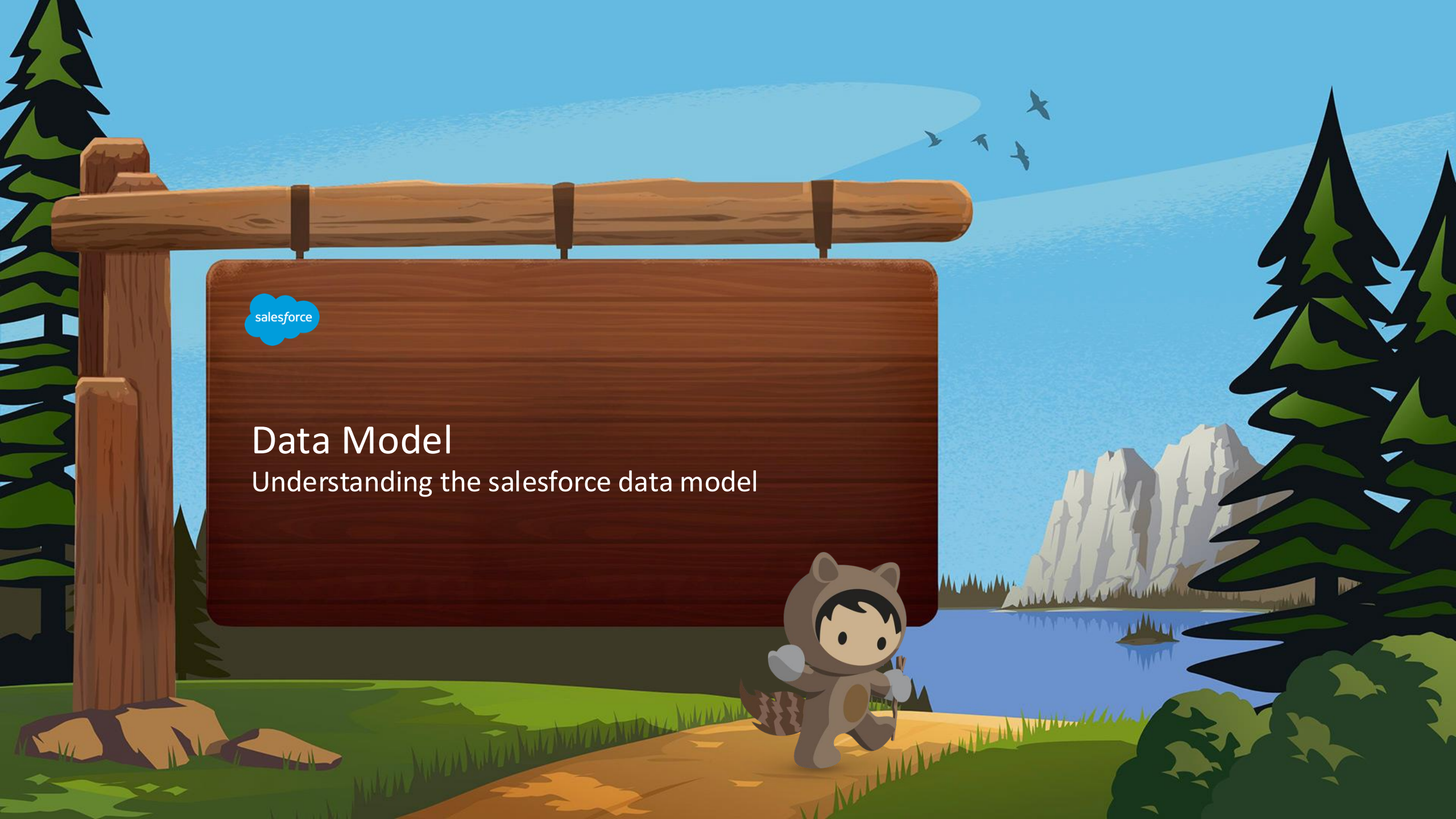
Ground Rules



Let's agree to some ground rules before we proceed

- Training will be delivered considering you do not know anything about Salesforce. So, even though you have prior experience keep your mind blank and open to learn more effectively.
- This is online live Training Session. Focus completely on session, even though recordings will be made available.
- Please make sure there is No background noise or disturbances.
- Questions will be addressed at the end of the session and will also setup a recurring meetings to discuss that week's question.
- **NO QUESTION IS SILLY.** So, you can ask same question again n again till you understand completely.
- **Phones on Silent.**
- **Complete all the assignments on time so you can understand next session.**
- **Do not forward invites unless you confirm with Trainer first.**





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Data Model

Understanding the salesforce data model

What is Data Modelling?

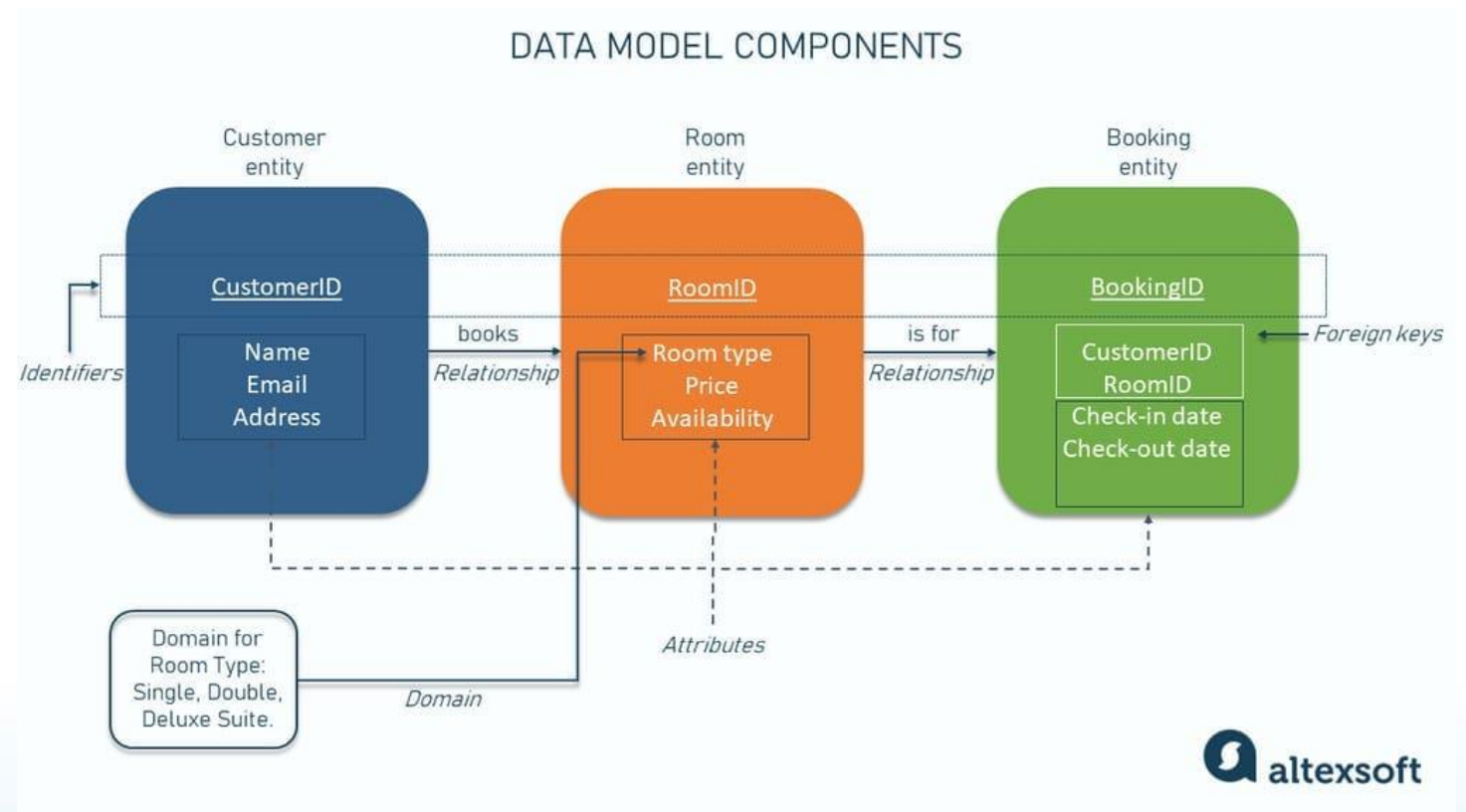
How your database is structured and entities related to each other. Helps understand data better.

What?

creating detailed diagrams or models of data. It deals with identifying what data is needed, where it comes from, how it moves, and how it should be structured.

GOAL

Create a visual data map describing data structure, data flow, key attributes and Relationships to align the business's core rules with data definitions, optimizing how data is stored and used for business activities.



Database Data Modelling & Design Options

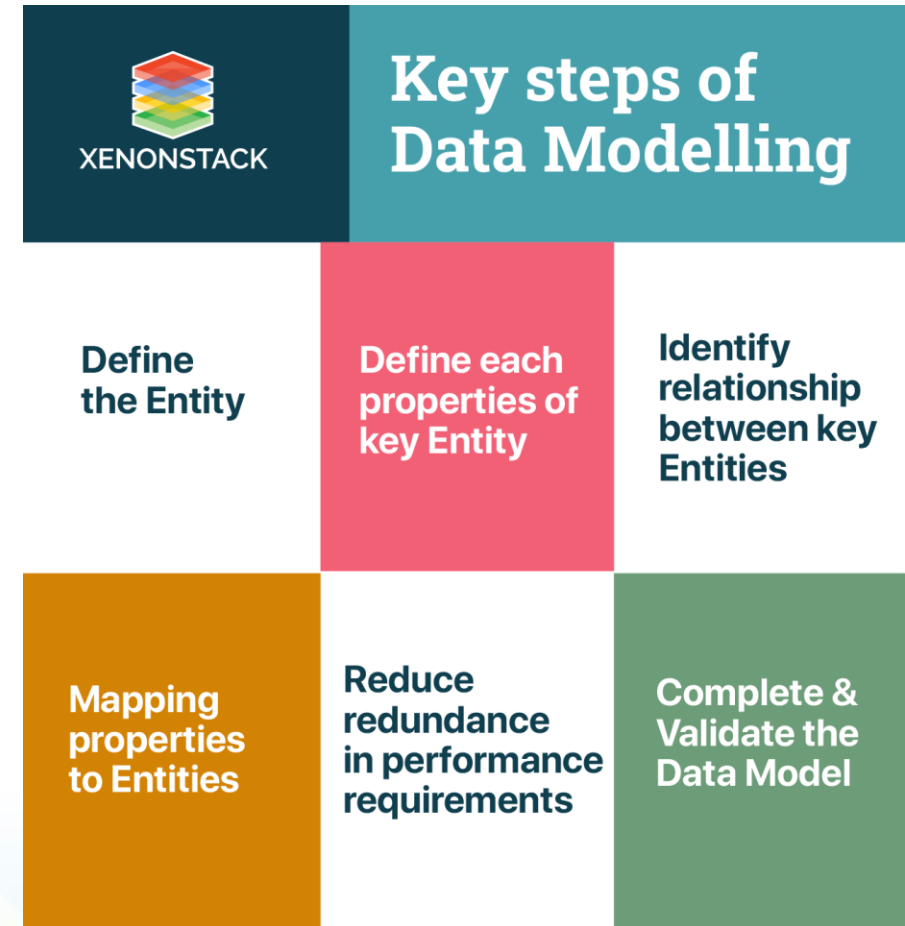
In modern software development there are three main ways of storing data:

Relational Databases (Entity models): These are made from tables (tabular data) with solutions including PostgreSQL and MySQL.

Object Databases (NoSQL): These are made up of key, value pairs and don't have a strict schema, several solutions include Firebase, MongoDB and Amazon DynamoDB. Instead of tables collections hold the documents.

Graph Databases (Tree): Graph databases are often seen within social networks such as Facebook, a graph database is composed of nodes and edges, several solutions include Neo4j, Dgraph and Grakn.ai. Used when your data is incredibly nested with many-to-many relationships.

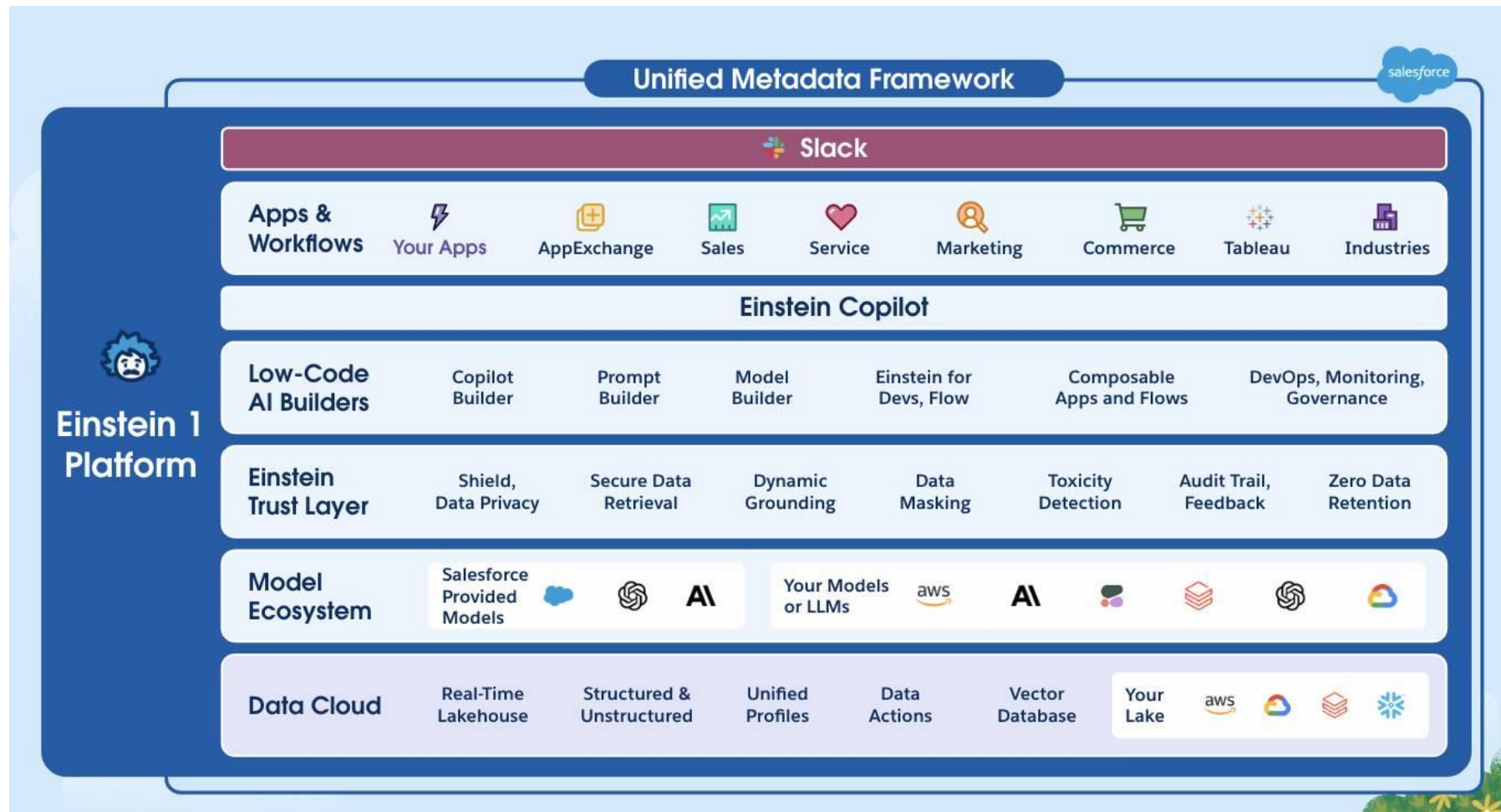
Data models are best created early on within a good data project.



What Is the Salesforce Architecture?



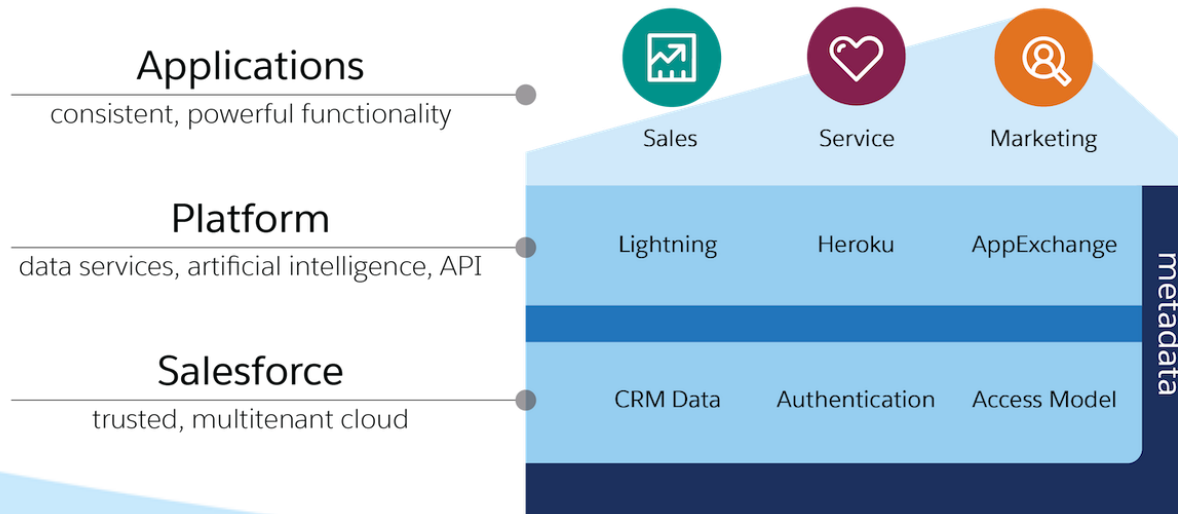
Series of layers providing strong security, reliability and foundation to build your apps in cloud.



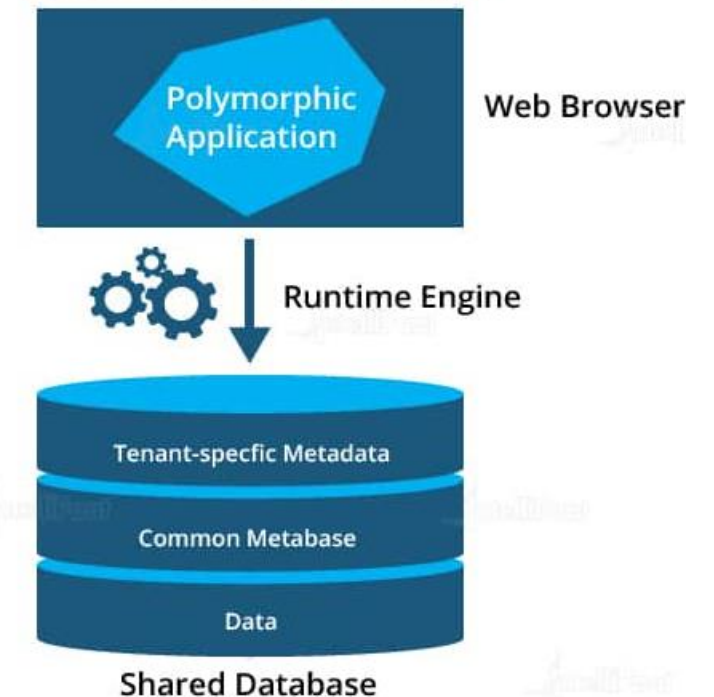
Architecture of Salesforce



- The Salesforce platform Layer is the foundation of services. It's powered by metadata and made up of different parts, like data services, artificial intelligence, and robust APIs for development.
- APIs help to communicate with other systems seamlessly.
- All the apps sit on top of the platform. and prebuilt offerings like Sales Cloud and Marketing Cloud Engagement, along with apps built using the platform, have consistent, powerful functionality.
- Salesforce platform uses a metadata-driven, multi-tenant development model.



Metadata Architecture





The Salesforce Database Architecture

The Salesforce CRM provides a robust database largely based on AWS (prior Oracle). The database also uses features of PostgreSQL and a few other languages for its functions.

- The Salesforce database is made of three functional components – Objects, Fields and Records
- The database in Salesforce is relational in structure, which means that the data is stored in **Tables**. And in Salesforce terms these **Tables** are called **Objects**.
- The tables or objects comprise **columns** referred as **Fields** that indicate the type.
- **Table rows** are the **records**
- Multiple objects can be associated with a single record, and these objects are interconnected using unique identifiers.
- Tables (Objects) are related through Relationships:
 - One-Many
 - Many-Many
 - Hierarchical
 - Self
- Types of objects that may be used in the database of Salesforce:
 - Standard Objects : OOTB provided by Salesforce and available for use. Like Account, Case, Opportunity.
 - Custom Objects: Created by users to cater the business needs.
 - External Objects: Map external systems data to Salesforce

Object $\Rightarrow$ Table or Entity
Row $\Rightarrow$ Record
Column $\Rightarrow$ Field or attribute



Key Concepts in Salesforce

Transactions:

- a transaction can be instantiated by code execution and/or a database operation.
- To maintain resources for all tenants, Salesforce imposes limits upon every tenant, calculated on a per transaction and per org basis.
- to ensure individual instances of code execution and database transactions don't monopolize shared compute resources Salesforce enforces Governor Limits.
- At Org-Level – limits are based on edition and feature types. Also, rolling calculation of API usage across all transaction.

Execution Context:

- Salesforce calculates transactional limits based on a concept of execution context
- based on the platform runtime engine, and all code available to the runtime engine for a particular tenant
- Execution context also controls the security of data.
- Context could be – running user (logged in user), Specified user, System Context

Database Manipulation

- When a transaction involves database manipulation, a built-in order of execution decides how the different configurations and code will behave.
- Save and Commit are two different operations in Database Manipulation.

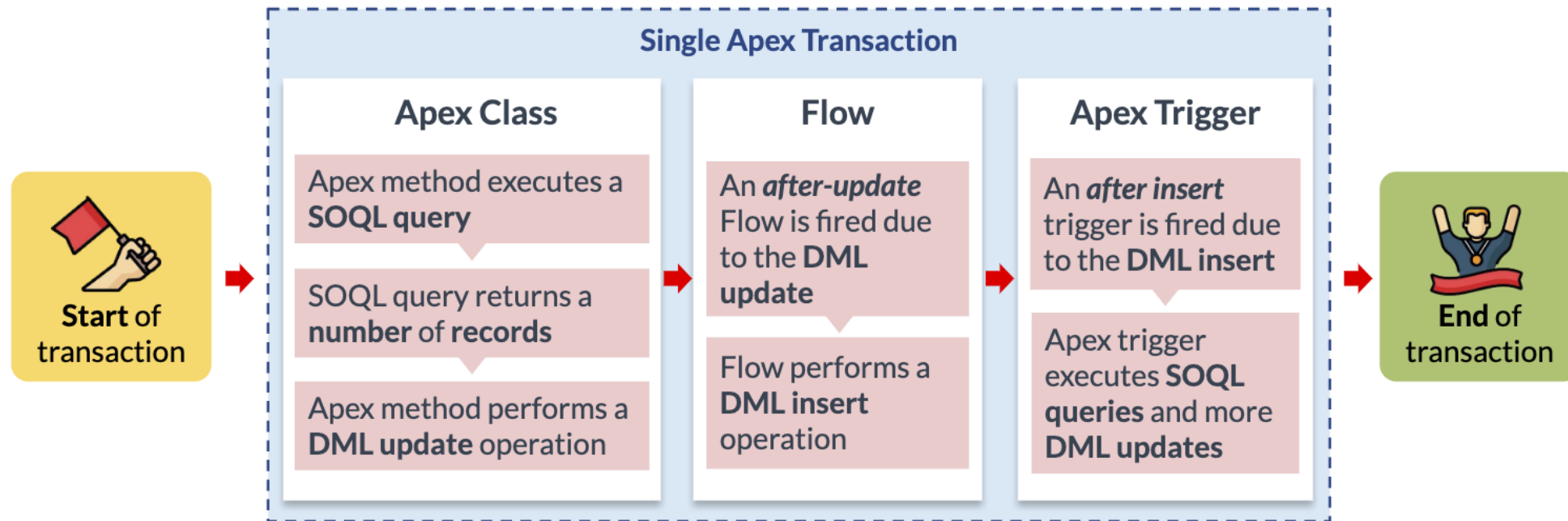


Transaction Scope



Transaction Example

A transaction is a **set of operations** that is executed and evaluated as a **single unit**. Each event that occurs in a single transaction are **bound** by their associated governor limits.





Metadata vs Data

When choosing how you customize and extend within a Salesforce org, it is important to understand what will be considered data and what will be considered metadata from the point of view of the Salesforce Platform

- This distinction will affect many aspects of the development lifecycle for your application, such as: whether an artifact will be copied over into sandbox development environments, how it can be migrated to other environments, whether it can be part of a package.
- The table below offers a quick comparison of the performance of data vs. metadata as it relates to key parts of the application lifecycle:

Behavior	Data	Metadata
Copied from production into sandbox environments	No*	Yes
Migrate by Metadata API	No	Yes**
Migrate by data load	Yes	No
Can be included in packages	No	Yes**
Counts against data storage limits	Yes	No



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Salesforce Core Objects

OOTB Standard Objects

Types of Objects

Objects are tables to store and retrieve data.

Types of objects

Objects are real life entities.
Used for tracking key data related to Entities

01

Standard Objects

Pre-defined by Salesforce.



02

Custom Objects

User defined tailored to fit business



03

External Object

Store data from 3rd party systems



04

Big Objects

Store & Manage LDV Data



Standard Objects

Anything that is provided by Salesforce as a part of the platform is called – standard or OOTB.

- Salesforce comes equipped with standard objects and fields, designed to capture essential business data.
- To collect additional data, you can customize your objects, fields, and, tabs in Salesforce.
- **ALWAYS OPT FOR OOTB FEATURES FIRST**
- Faster to deploy, easy to configure and upgrade. Low maintenance.
- Objects are made up of **fields**. **Fields** store the specific information that must be filled in for entity (for example, First Name).
- Standard fields are a set of fields that come already set up

Salesforce Standard Object	Standard Fields
Account	Standard Account Fields
Contact	Standard Contact Fields
Lead	Standard Lead Fields
Opportunity	Standard Opportunity Fields

These are key Salesforce Objects which are frequently used. There are other platform objects as well.

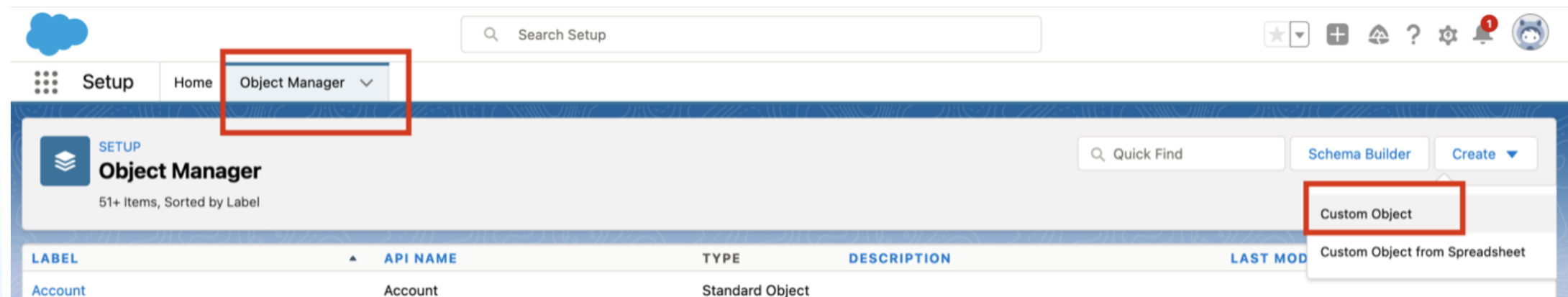


OOTB – Provided as a part of Product. In this case provided by Salesforce when you create new org.

Custom Objects

User defined to store information that's specific to your company or industry.

- The organization's data will always not fit into the existing standard objects. Use Custom Objects for this purpose.
- The objects created by users are called custom objects. Custom objects are usually identified by a __c suffix.
- Store different types of information in any organization. Create custom objects according to requirements and manage org data easily.
- Custom objects already have some standard fields, like Name and CreatedBy, and allow you to add other fields that are more specific to your implementation.



Object Definition

At the time of creating a Custom Object, on the Custom Object Definition page, Salesforce gives you multiple options





Custom Object Properties

- ☐ **Allow reports:** If we check this check box, then only these objects are available to create reports.
- ☐ **Allow Activities:** If we check this check box, then we can create activities on this object.
- ☐ **Track Field History:** Track audit history of changes for this object. We can track up to 20 fields for a single object.
- ☐ **In development:** If we check this check box, this object is still in development mode. This object is not available for deployments.
- ☐ **Deployed:** After selecting this check box then only it will be available for deployment.
- ☐ **Create Tab:** You can even create a tab later or immediately. Tab allows to access records using User interface.

These properties cannot be edited for the Standard objects.



Other Objects



Salesforce supports objects to help you manage large data efficiently.

External Objects

External objects are like custom objects.

External object record data is stored outside of your Salesforce organization.

They allow us to map the data that's stored outside your Salesforce organization.

For example, You have data that is stored in the ERP system or any other place.

Big Objects

Manage billions of records of data without compromising performance.

Store and manage a massive amount of data in Salesforce.

Archive or bring massive datasets from outside systems into a Big object to get a full view of your customers.

Provides consistence performance whether you have 1 million or more records.

Can't use standard features of Salesforce like triggers, flows, and processes

Big objects are usually identified by a __b suffix.



Key Terms

CRM

Boost your success with Salesforce Customer Relationship Management (CRM), which allows you to manage relationships with your customers and prospects and track all of your interactions.

SEARCH

Make searching data easy with powerful search features, including filtering and customizable search results layouts.

ACCOUNTS

Accounts are companies, entities, or organizations you do business with. Accounts can be businesses or individuals. Use accounts to track important information, including name, address, phone number, and related information like opportunities, activities, cases, notes, and more.

CONTACTS

Contacts are the people who work at (or are associated to) accounts. Using contacts in Salesforce, you can track key details such as phone numbers, addresses, titles, and more. You can also indicate a contact's role in a particular opportunity, such as decision maker or influencer.

EMAIL TEMPLATES

Make it easy to stay in touch with your accounts and contacts by using standard and customizable email templates.

IMPORT WIZARD

Easily import data into Salesforce. Map your information to leads, contacts, accounts, and more.

WEB-TO-LEAD

Create a simple online form for prospects to ask questions or sign up for events and free trials, and automatically create leads from submissions.

LEADS

Leads are prospects or potential opportunities, such as a person you met at a conference who expressed interest, or someone who filled out a form on your company's website.

OPPORTUNITIES

Opportunities are leads you've qualified to buy. Opportunities are your pending deals which fill your pipeline, which is where you track all of your deals in progress.

TASKS

Tasks are your to-do items which require follow-up. They can be assigned to you or someone else, and are related to leads, contacts, accounts, or other records.

ACTIVITIES

Activities are where you log your calls and activities with prospects and customers, allowing you to keep a complete history of your interactions.

EVENTS

Events are meetings you've arranged with your prospects and customers, all tracked in Salesforce.

REPORTS

Summarize and analyze your data with reports, which you can display, share, collaborate on, export, and print. Use standard reports or create custom reports. You can subtotal and filter your data to see exactly what you need.

DASHBOARDS

Access visualizations of your metrics and key performance indicators, through sets of charts and graphs. Visualize multiple reports on a single dashboard, using pie charts, bar charts, and more.



Resources

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- [Get to Know Data Modeling](#)
- [Custom Objects: Quick Look](#)

thank you

